Noise Reduction for Flow Past a Blunt-Based Thick Profile by Modifying Geometry

Cedric Kouakou*, Benjamin Milner*, Phani Panchangnula*, and Antonio Vicedo-Pina*
Université de Toulouse, ISAE-Supaero, 31400, Toulouse

In this paper, an analysis of the Kármán vortex shedding was performed on a blunt-based thick profile both experimentally and numerically. Vortex shedding occurs when the fluid moves past a blunt trailing edge, and vortexes at the back of the body and detach periodically. This causes an unsteady periodic interaction between the flows moving on either side of the profile which is the main driver for the aeroacoustic noise created. By first characterizing the unsteady flow past the blunt-based body, the geometry was modified by adding one, then two separator plates to limit interaction and see the effect on the vortex shedding frequency. By decreasing the frequency, the aeroacoustic noise also decreases. This was done experimentally in the ISAE-Supaero wind tunnel, and also numerically in Star CCM+. The chosen velocities for this experiment were 10, 15, and 20 m/s. The results show that the middle separator plate increases the vortex shedding frequency by 12-18%, while the use of two separator plates reduces this frequency by 10%.

I. Nomenclature

D	=	Blunt Body Thickness
St	=	Strouhal number
Re	=	Reynolds number
TE	=	Trailing edge
FFT	=	Fast Fourier Transform
U	=	Freestream Velocity

*MAE Students, ISAE-Supaéro, Department of Aerodynamics, Energetics and Propulsion.

II. Introduction

In domain of aerospace, noise is an omnipresent problem. Aircraft flying through the air generate a lot of noise whether it be from the engines or the huge amount of air it needs to displace to fly. Aircraft are so loud that regulations have been put in place to limit the amount of noise they can produce. This is why aeroacoustics has become a very prominent field of research to determine methods to decrease the amount of noise created by a flying aircraft.

The main purpose of this project is to decrease the noise generated by a blunt-based thick profile through the mitigation of vortex shedding by modifying trailing edge geometry. This investigation has been performed with both an experimental and computational campaign. Ideally, when constructing an airfoil, the trailing edge should be sharp and have zero angles so that the upper and lower surface flows meet and no interactions occur. However, in reality, this is never the case and airfoils always have some bluntness at the trailing edge. This small thickness at the Reynolds number at which aircraft fly causes a significant amount of noise. While this noise comes from a variety of sources, one key source is vortex shedding.

Vortex shedding occurs when a fluid flows past an object, creating alternating swirling patterns or vortices in the wake of the object. These vortices are shed from the object at a certain frequency and lead to the generation of noise. This shedding happens at a specific frequency known as the Strouhal frequency, which is determined by the object's shape, size, and speed of the fluid. The shedding vortices create a periodic fluctuation in the pressure. This variation in pressure then leads to noise generation. The following diagram illustrates the concept:

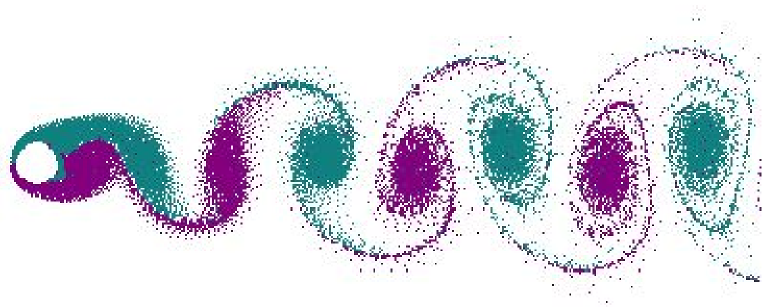

Fig. 1 Graphic demonstrating the vortex shedding past a cylinder

By modifying trailing edge geometry, we can influence the interaction of the vortices generated by either side of the body. Furthermore, because we know the noise is generated by the frequency of pressure fluctuations, by decreasing the frequency of these oscillations, the noise will decrease as well.

III. Objectives

It is our objective to analyze the characteristics of the wake flow of a blunt body at different operating speed. With this, we intend to obtain better understanding of the vortex shedding phenomena and be able to modify via geometrical modifications it in order to reduce its frequency.

With this, it is expected to reduce the noise discomfort produced by the vortex shedding by: 1) reducing the shedding frequency and thus the noise frequency and 2) reducing the pressure difference generated by the vortex which is responsible for the intensity of the noise.

IV. Current Research

The problem of the thick profile has been widely studied with the context that, when manufacturing airfoils and other aerodynamic elements, the impossibility of manufacturing a true sharp geometry causes the apparition of the vortex shedding. Fanning [1] and Boldman [2] characterized the phenomena and established some early bases in the past century by performing analysis at different Re and characterising part of the Re-St curve for the baseline geometry.

In the present, research like the one developed by Taherian et al. [3] show how the study trends aim towards the direct application of flow control strategies into thick airfoils in an attempt to reduce the vortex shedding frequencies that are observed. This validates therefore the purpose of our paper.

V. Experimental Methodology

A. Wind Tunnel setup and Test Model

The measurements were made in a subsonic wind tunnel with test section dimensions (45x45) cm^2 with the following flow velocities: 10, 15, and 20 m/s. These velocities were determined using a pressure probe located in the convergent section of the wind tunnel. The turbulence intensity of the tunnel is approximately 1.5%. The calibration curve was given and has an uncertainty of 0.3m/s. A temperature was also noted.

The test model for this experiment was a blunt body, with a thick trailing edge. Its dimensions were 45cm in width, 25cm in length, and the trailing edge was 2.5cm in thickness. The leading edge of the profile was 14cm into the test section, which we used as a reference for the measurements we took. The body was made from 3D-printed material and the sides were directly screwed onto the sides of the test section. The tripping wire was made from 10 thin layers of painter's tape. The plates used to modify the geometry were made from cardboard, and covered in a layer of aluminum tape to decrease the surface roughness. The following is a side view of the test setup with dimensions in red:

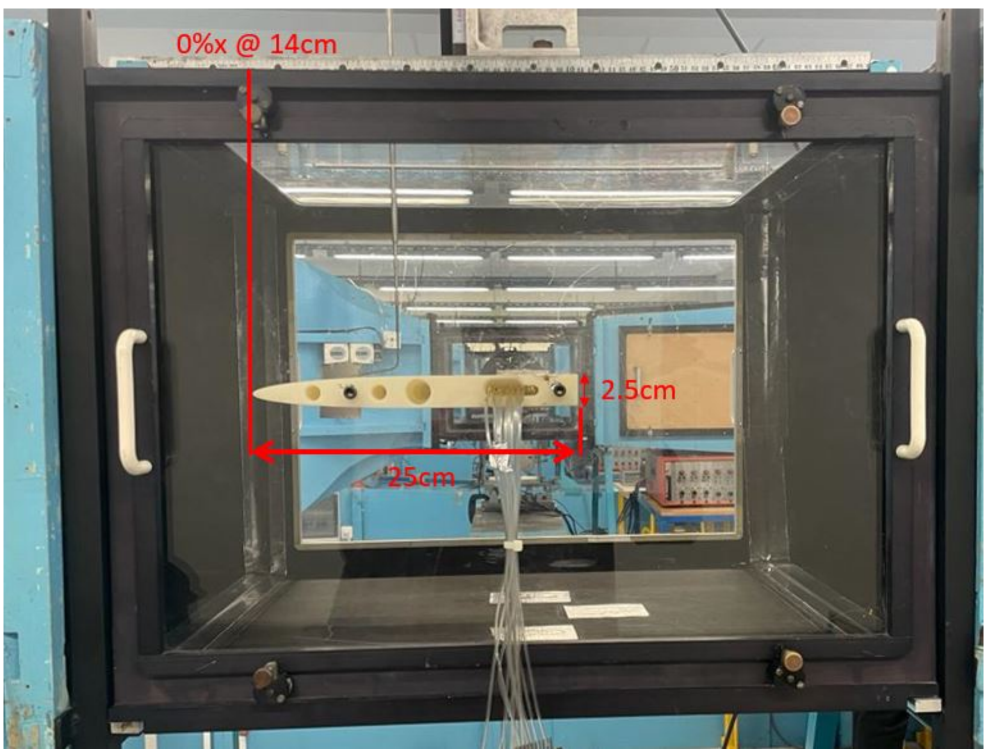

Fig. 2 Side-view of experimental setup

In order to regulate the test conditions effectively, we positioned a tripping wire on the upper surface of the profile, at approximately 5% of the chord. This deliberate placement induced a transition to a turbulent regime, aiming to simulate the flow characteristics observed in actual wings where laminar flow is absent. Moreover, this setup ensured a relatively uniform boundary layer across all conducted tests, enabling consistent results. This setup was mirrored in the CFD.

B. Experimental Methods

Several experimental methods were used in this project. Due to the nature of the study, in order to characterize the wake flow, we needed to understand the general characteristics of the wake flow. To do this we used a forked pitot probe allowing us to measure total and static pressure using the same probe. With these probes, we are able to determine the time-averaged velocity profile at certain distances from the trailing edge. Measurements were taken in the wake at 3 fixed distances from the trailing edge and for a whole sweep on the vertical axis. This allowed us to see the boundary between the free stream and the wake after the trailing edge, informing the vortex shedding study.

Since this project's main objective was to analyze the vortex shedding past a blunt-based body, our main focus is on frequency. While a pitot probe is a powerful tool for determining flow speeds, it is not responsive to rapid changes. Due to the way a pitot tube functions, when high-frequency variations occur, it takes time for the probe to react, due to the inertial nature of the probe. Therefore, instead of a pitot probe we use a hot-wire probe (HWP) to measure changes in the wake flow. A HWP consists of a thin wire, typically made of a high-resistance material such as platinum, which is heated by passing an electric current through it. The wire is maintained at a constant temperature above the ambient temperature of the surrounding fluid. When the air moves past the wire, it cools it, which changes the resistance of the wire. Then using a Wheatstone bridge, this change of resistance is converted into a voltage read-out, which can then be converted into velocity. Using prior knowledge, we estimated our shedding frequency to approximately 250Hz, therefore to obtain accurate data, we collected 10000 points, at 6000 samples per second, meaning we had a resolution of 0.6 Hz for each point in the wake we measured. From the information gained in the time average wake study, we determined the relevant locations for the hot-wire study; these being at 1-D* 2-D*, and 3-D* away from the trailing edge of the model, for an entire sweep of the vertical axis. Two points were collected in the free stream starting at the top of the tunnel section. After, measurements were taken descending through the tunnel by 5mm each time. Finally, two more points in the free stream were collected at the bottom.

Three geometries were selected for the study. The first is the unaltered model. The second geometry was an altered version of the model, with a plate of length 1-D (2.5cm), being placed in the middle of the trailing edge. The theory is that by limiting the interaction, the mixing is delayed, and thus the frequency will decrease. The third geometry was the model with a plate of 1-D placed on the top and bottom of the trailing edge, which created a cavity behind the model, thus delaying the interaction between vortices, and in theory, decreasing noise. However, due to time constraints, only geometries 1 and 2 have been tested experimentally enough to provide results. Below, see the diagrams of the three geometries:

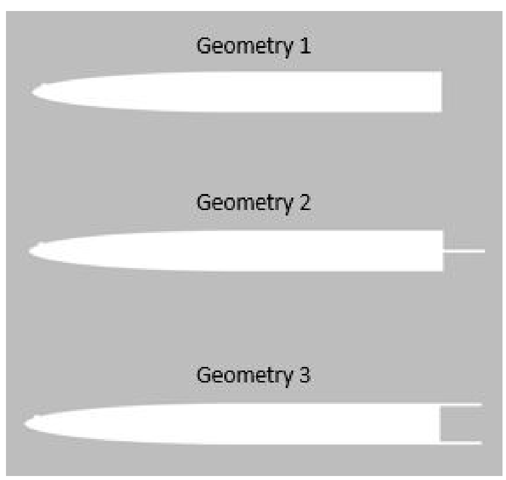

Fig. 3 2D models of the three geometries selected for this study. A tripping "bump" has been included to show where the tripping wire was placed

*D refers to the thickness of the trailing edge, that being 2.5cm. Therefore, 1-D is 2.5cm away from the trailing edge.

Table 1 CFD Mesh Parameters

Parameter	Value
Domain Size	5m x 5m
Cell Base size	0.03 m
Cone refinement	5 percent of base
PL Thickness	7.833
PL Stretching	.1
PL No. of Layers	34
Time Step Selected	0.0001 s
Velocities	10-15-20 m/s

VI. Numerical Model

In order to perform simulations of the case, a computational domain had to be created. We have selected a 2D configuration. The criteria of having a 20 chords length in all directions has been followed, leaving us with a 5x5m box that contains the study geometry in the centre. Since transition to turbulent BL will only be forced on the upper surface, a geometrical imperfection has been modelled into the body, as seen in Fig 3. The inlet and outlet characterizing parameters are the inlet velocity and static pressure respectively. The upper and lower surface have been set to bee mirror planes. The meshing process consisted on firstly determining the prismatic layer parameters targeting a y+ of 1, and secondly refining the wake region so as to obtain better resolution in that area. The main parameters of the mesh are depicted in Table 1 , and some detail images are shown in Fig 4.

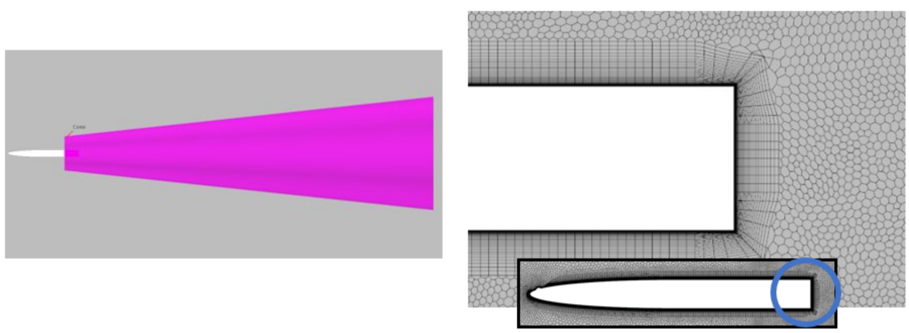

Fig. 4 CFD Mesh Volumetric control for wake region and Mesh Focusing on trailing edge of the Blunt Body.

First, a Mesh Independence study is performed on the blunt body with any modifications to the Geometry. The initial Mesh selected was with 100,000 elements approximately. Then The base cell is size is changed two more mesh are use for grid independence study with 250K and 500K elements. The simulation are are first started with steady state model to have a first initial solution for the unsteady simulations. After that the unsteady solver is used selected and the simulations were performed for 2000 time steps. The results of these mesh study is depicted in Table 2. Based on the data the final mesh selected is with 250k elements. To judge the convergence of the simulations, repeating oscillations of the residuals is considered.

Table 2 Mesh Independence Study

Number of Elements	Cd Max	% variation	Frequency (Hz)	% Variation
100 K	1.22	-	454	-
250K	1.36	11.4	400	11.9
500K	1.37	0.75	416	4

Table 3 Mesh study results for different turbulence and transition models

Model	Frequency	% Variation
Experimental	211	-
Spallart -Allmaras	188.6	11.87
K- Epsilon	188.6	11.87
K-Omega SST , Gamma Transition	196.1	7.06
K-Omega SST, Gamma Re Theta Transition	196.1	7.06

Once we have the final mesh selected, another study was done to select the Turbulence and Transition models. The frequency and its variation for different models is depicted in Table 3 . Based on the data K - Omega Turbulence model with Gamma Re Theta Transition model is selected as the model to be used for the simulations with modified geometries.

VII. Results and Discussion

A. Experimental Results

Using the data from the HWP, we were able to complete a fast Fourier transform (FFT) for each of the points where measurements were taken. To increase the accuracy of the FFT the technique of "Windowing", a signal-averaging technique, was used. After completing an FFT, by looking at the peak, we are able to determine the strongest frequency of the signal. Furthermore, to guarantee that a peak was not just noise, we set a minimum threshold of 0.4 for the value of amplitude. Taking these peaks and plotting them as a function of their vertical position, we are able to see the "frequency" profile in the wake. Below see an example of the methodology used to derive the frequency profile plots:

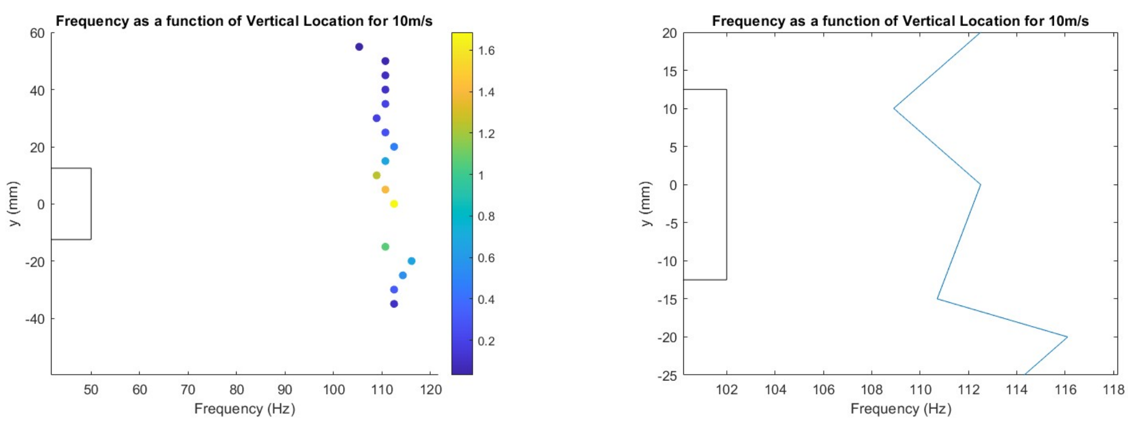

Fig. 5 Plots demonstrating frequency profile in the wake for 10m/s at 1D. The left plot depicts the frequency at which the max value occurs in the FFT for each point, and its magnitude is represented by the color of the points. The right plot is the frequency profile after the points from the left plot with an amplitude of 0.4 or lower had been removed. The trailing edge in the profile has been added for visualization purposes.

Below, please see the frequency profiles of this analysis for the two geometries, at the three speeds 10, 15, 20m/s.

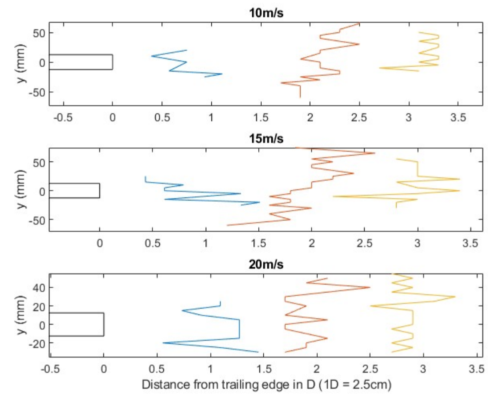

Fig. 6 Frequency profile of unaltered geometry

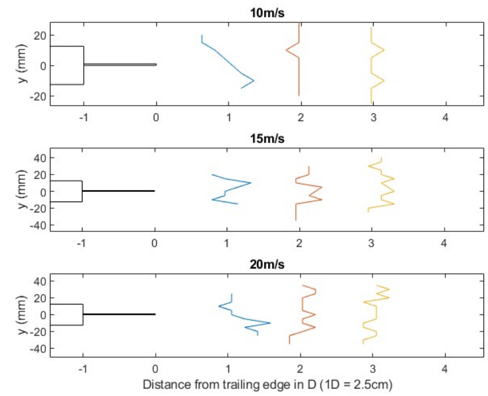

Fig. 7 Frequency profile of geometry 2

Note: These plots are of the frequency profiles, where the further to the right a point is, the higher the frequency measured by the HWP

Immediately we see, no clear pattern emerged for any of the horizontal locations and most of the results appear quite noisy. However, two conclusions can be drawn from these plots. The first is the fact that at a certain distance from the trailing edge in the wake, the frequency of vortex shedding appears to be constant for any point in the vertical direction at that distance. The second conclusion comes from comparing the two figures above. By adding a plate to limit interactions between the vortexes, we seem to have stabilized the frequency profiles. That is to say, the profiles are more uniform through the sweep. However, this uniformity does not provide information regarding the magnitudes of the frequencies or whether the plate genuinely reduces the frequency of vortex shedding. To comprehensively assess the impact of adding the separator plate, the following figure plots all of these profiles together:

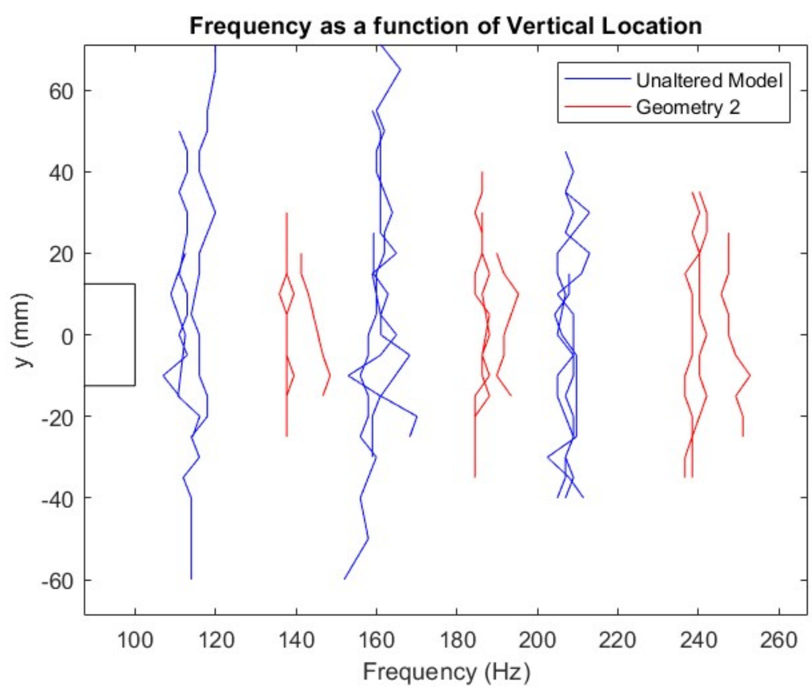

Fig. 8 Frequency profile of unaltered geometry

From this plot, we see that by adding the separator plate to make "Geometry 2", we actually increase the vortex shedding frequency by 20-30Hz at each distance, with the greatest difference at 20m/s (shown by the two groupings of profiles furthest to the right). The plate, therefore, had the opposite of the predicted effect and increased the frequency of vortex shedding, increasing the noise generated. The sweep-averaged values of the frequencies are depicted in Table 4.

Table 4 Shedding frequency values of the different configurations for different velocities obtained from the experimental tests.

Frequency	Clean (Hz)	Middle (Hz)	Variation (%)
10 m/s	115	142	+23 %
15 m/s	159	187	+17 %
20 m/s	211	246	+16 %

B. Numerical Results

Nine cases (3 geometries for 3 velocities) were executed. With the information obtained from these results, not only the shedding frequencies have been calculated, but also we have focused on the flow morphology.

We have performed an analysis of the shedding frequency by analyzing the variation of velocity with respect to the time registered in the probes that have been presented in the previous section. Since the simulations are purely periodical once they reach stability, the frequency has been obtained by measuring the time between consecutive peaks, without need for an FFT. Figure 9 depicts said evolution of velocity.

Table 5 Shedding frequency values of the different configurations for different velocities obtained from the CFD solution. Variations of frequencies with respect the clean geometry.

Frequency	Clean (Hz)	Middle (Hz)	Variation (%)	Double (Hz)	Variation (%)
10 m/s	99	111	+12.1 %	89	-10.1 %
15 m/s	149	167	+12.1 %	135	-9.39 %
20 m/s	196	222	+13.3 %	175	-10.7 %

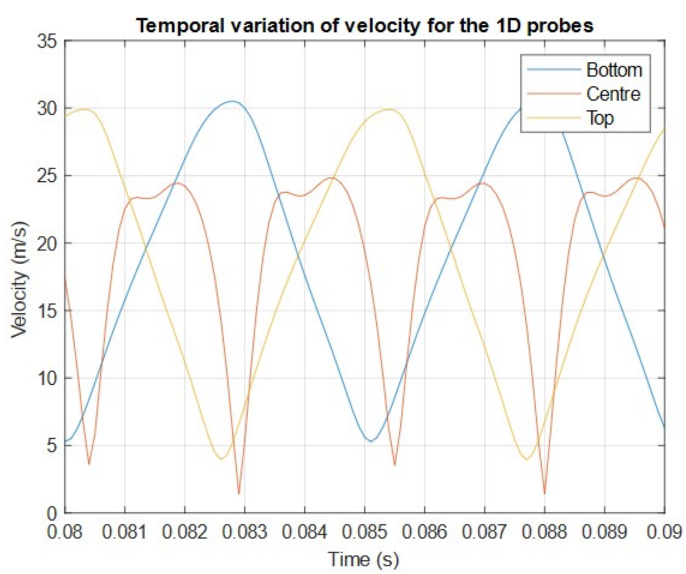

Fig. 9 Temporal evolution of velocity for the three 1D probes.

As seen in Table 5, the middle plate configuration has caused an increase in the frequency of 12% approximately for all of the studied velocities. On the other hand, the double plate geometry achieves a 10% reduction of the shedding frequency in the whole domain of study.

In order to asses the shape and evolution of the generated vortexes, vorticity contour plots of the wake behind the blunt body have been performed for the three geometries at a velocity of 20 m/s. We have noticed how the vortical structures in the clean configuration present higher vorticity values, and thus have higher spinning motion. Moreover, we saw how the wake quickly evolves by moving the vortexes away from the middle symmetry plane, thus helping in the dissipation process. On the other hand, the wake presented by the modified geometries present lower vorticity structures, which however manage to propagate further down the wake without being dissipated. The distance between vortexes in the double plate configuration is appreciably bigger than the one in the middle plate, result of the difference in the shedding frequency.

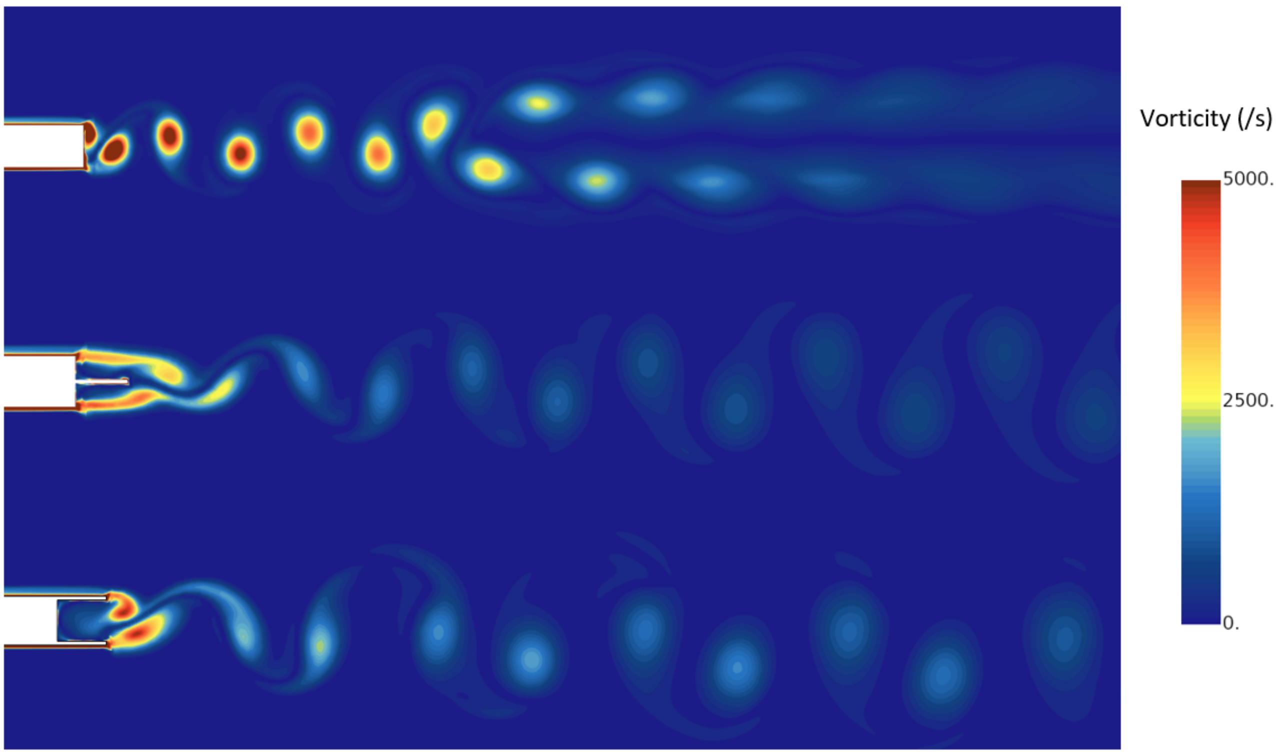

Fig. 10 Vorticity contour plots for the wake section of the domain for different geomety cases at 20 m/s.

It is also interesting to discuss the difference in minimum pressure values that were obtained. While the emitted sound frequency depends on the shedding frequency, the sound intensity will be determined by the static pressure difference between the vortex and the free stream. The clean case presents the higher difference, while the middle plate shows the lowest variation. The double plate lays in between, closer to the middle plate one. In terms of acoustic comfort, a high frequency low intensity sound can result more disturbing than other sound with higher frequency but lower intensity. This is why the double plate is regarded as more advantageous in terms of noise comfort improvement. Table 6 shows this difference for the cases at 20 m/s.

Table 6 Maximum pressure differences between free stream and vortex region for the three geometries at 20 m/s.

Pressure difference (Pa)	Clean	Middle	Double
20 m/s	716	350	428

C. Agreement Between Experimental and Computational Results

Once all of the results were obtained, we carried an agreement analysis to check the matching between the experimental and computational results. In order to do so, the evolution of the Strouhal number against the Reynolds number for the analyzed domain has been plotted for all of the configurations. We appreciated in both Figure 11 and Figure 12 how the experimental values for St are higher in the whole studied domain, with a decreasing tendency that is accentuated in the case with a middle plate. On the other hand, the computational results bear lower St values, which remain practically constant in all of the domain. Therefore, we can state that for the computational setup that we used, frequency is directly proportional to the inlet velocity as seen in Equation 1. The results can also be used to compare the performances of the three studied geometries: lower St values bear lower shedding frequencies and vice versa.

Table 7 quantifies the variations in the St presented previously. We appreciated a relatively high disagreement between the results. Higher differences were registered for low Re numbers, being the values variation lower with the increase of Re. The literature used as reference [1] present results whose error between experimental and computational results range from 15% to 20%. Moreover, their values for the clean case at Re of the same order of magnitude provide absolute St values in the range of 0.2 to 0.3, trends that we observed in our analysis as well. Thus, we have concluded that our data lies within the range of acceptable error.

The general tendencies in the change of frequency observed in the experimental setup have been replicated by the CFD solutions. Thus, we can state that the used computational model can predict these trends as a function of the geometry. This validates the solutions provided by the computations on the double plate configuration, geometry from which we lacked experimental data, allowing us to state that said geometry decreases the vortex shedding frequency.

$$f = \frac{St}{D} U = K\, U \tag{1}$$

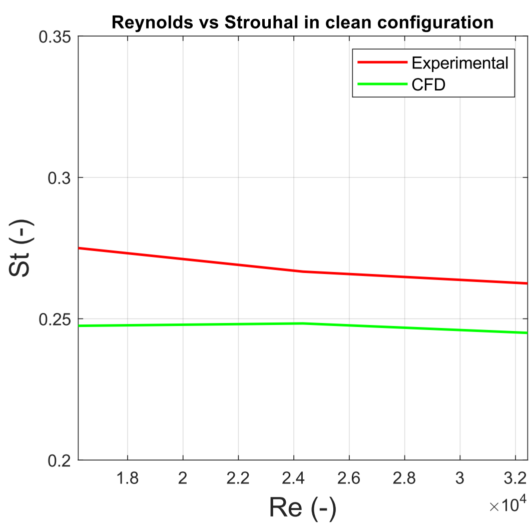

Fig. 11 St vs Re for clean configuration

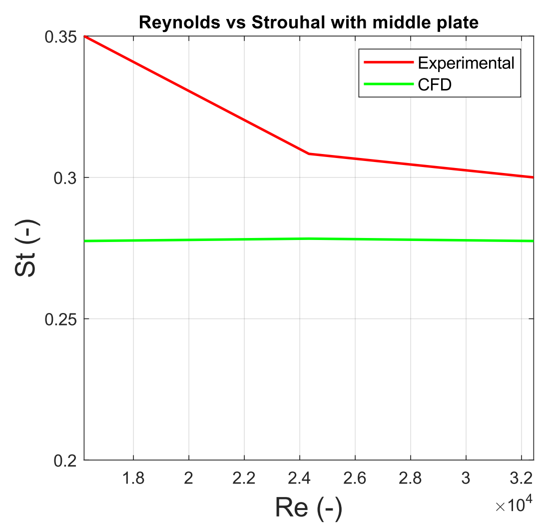

Fig. 12 St vs Re for middle plate configuration

Table 7 Error between the St values provided by the experimental and CFD results for the clean and middle plate configuration and various Re

Strouhal Error (%)		**Clean**	**Middle**
Re (-) (x 10^4)	**1.6**	10.0 %	20.7 %
	2.4	6.87 %	9.73 %
	3.2	6.67 %	7.50 %

VIII. Conclusion

The presented work showed the effects that the modification of the TE geometry of a blunt body had in the vortex shedding frequency, and thus in the noise properties.

We have observed an increase in this frequency for the middle plate geometry, both in experimental and numerical analysis. Nevertheless, a decrease in the frequency has been observed in the double plate configuration. While we have concluded the CFD results to be reliable, an experimental analysis of said geometry is suggested as an extension of the study to be able to contrast and validate the data obtained.

On the other hand, the use of acoustic measurement equipment will be used in next tests together with the hot wire to validate the hypothesis that correlates the shedding frequency with the acoustic properties of the flow.

The use of this kind of geometry in aerodynamic elements such as wings will help reduce the aircraft noise and thus reduce this kind of pollution. Its interest lays within the fact that the geometrical modification of the TE of the wing is a passive measure that imposes no penalty on the generated drag.

References

[1] FANNING, A. E., and MUELLER, T. J., "Numerical and Experimental Investigation of the Oscillating Wake of a Blunt-Based Body," *AIAA Journal*, Vol. 11, No. 11, 1973, pp. 1486–1491. https://doi.org/10.2514/3.50615, URL https://doi.org/10.2514/3.50615.

[2] Boldman, D. R., Brinich, P. F., and Goldstein, M. E., "Vortex shedding from a blunt trailing edge with equal and unequal external mean velocities," *Journal of Fluid Mechanics*, Vol. 75, No. 4, 1976, pp. 721–735. https://doi.org/10.1017/s0022112076000499, URL https://doi.org/10.1017/s0022112076000499.

[3] Taherian, G., Nili-Ahmadabadi, M., Karimi, M. H., and Tavakoli, M. R., "Flow visualization over a thick blunt trailing-edge airfoil with base cavity at low Reynolds numbers using PIV technique," *Journal of Visualization*, Vol. 20, No. 4, 2016, pp. 695–710. https://doi.org/10.1007/s12650-016-0405-3, URL https://doi.org/10.1007/s12650-016-0405-3.